

Basic Principles of Medicine II

Module: Nervous System and Behavioral Sciences

Lecture No: NB 01

Neuroembryology

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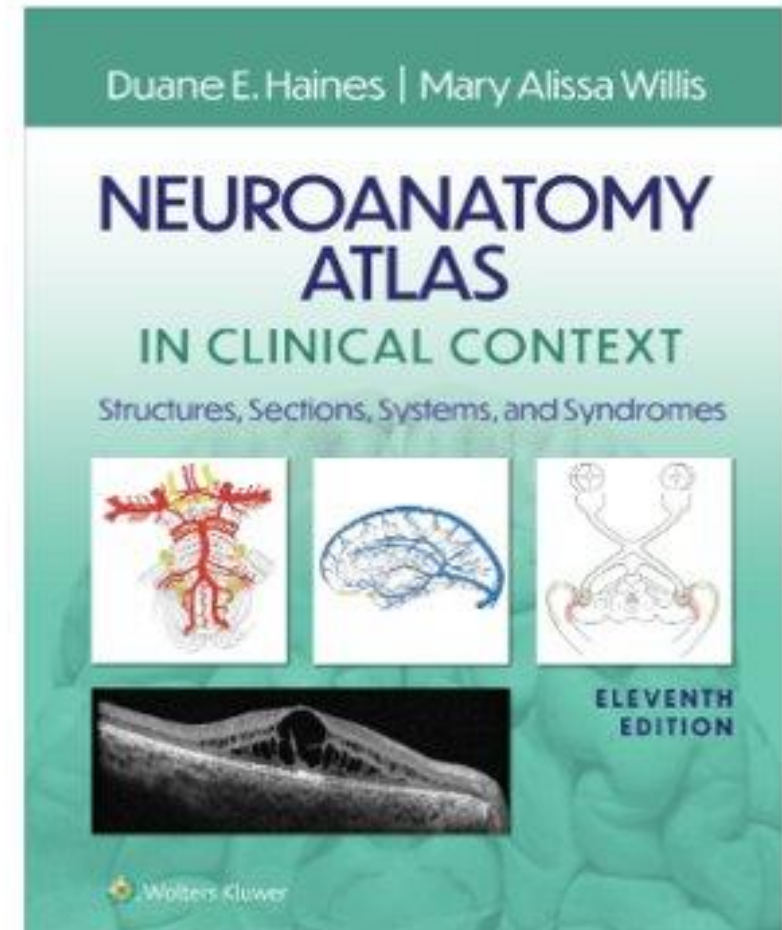
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Blue Slides

- Slides with bluish backgrounds may provide clarifications but are not intended as targets for memorization.

Neurobehavioral Module: General Overview

- Basic Principles
 - Neuroembryology
 - Structure
 - Function
 - Cellular Neuroscience
 - Blood supply
- Neurological examination
 - Sensory systems
 - Motor systems
 - Cranial nerves
 - Complex brain function
 - Will mesh with topics in Behavioral Sciences
- Basic Principles



Introduction and Approach to NSCI in BPM 501



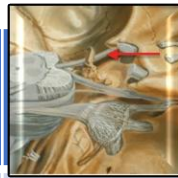
BLOCK 1

Fundamentals of Neuroscience

Head, Neck and Cranial Nerves

Histology of the Eyes and Ears

BPM 2 Exam 3



BLOCK 2

Advanced Clinical Neuroscience

Sensory and Motor Systems

Clinical Relationships

BPM 2 Exam 4



BLOCK 3

Limbic/Affective Systems
Hypothalamic Systems

Psychotic and Disruptive
Behavioral Disorders

Disorders of Cognition
Addiction Medicine

Lab Exam
OSPE
BPM 2 Exam 5

Lecture Objectives

- **Students would be able to:**
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0542** Describe the processes underlying neurulation.
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0543** Describe the formation of the vesicles of the brain.
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0544** Describe the morphogenesis of the forebrain.
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0545** Describe the morphogenesis of the spinal cord.
- **SOM.MK.I.BPM2.1.NB.1.NSCI.0546** Describe the morphogenesis of the brainstem.
- **SOM.MK.I.BPM2.1.NB.4.NSCI.0547** Describe structural abnormalities in the developing nervous system.

Recommended Reading

- [Development of the Nervous System | Essential Neuroscience, 4e | Medical Education | Health Library \(oclc.org\)](#) Siegel & Sapru

Introduction

- Hans Spemann - German Embryologist
- Awarded a noble prize in 1935 for his student's discovery of embryonic induction.

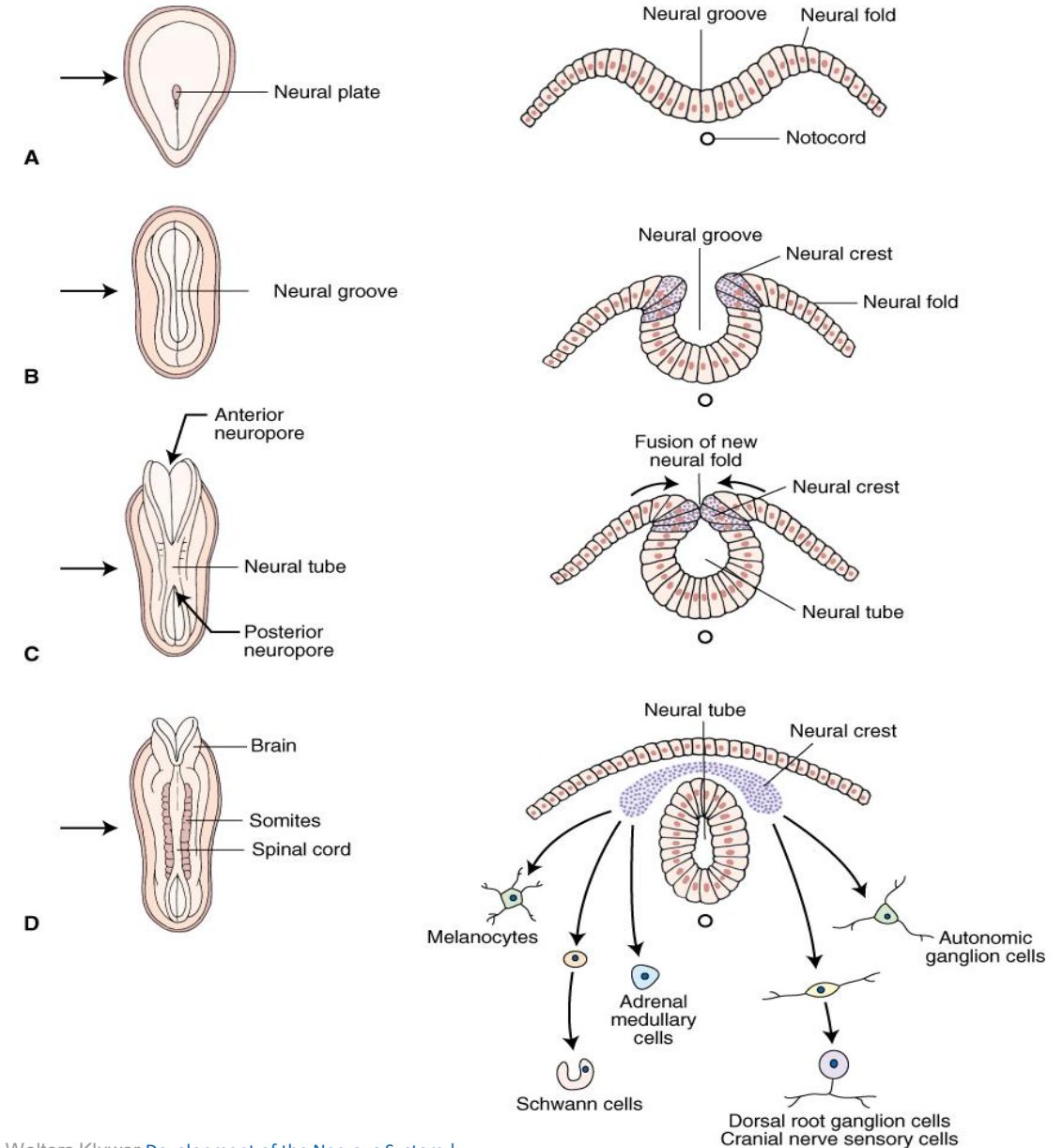


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Early embryonic Development of the CNS - Wk. 3-4

- The Nervous System develops from **ectoderm**.
- The **notochord** induces the development of the neural plate.
- Convergent extension of the neural plate causes the formation of the **neural folds**.
- The **neural groove** is formed along the longitudinal axis as the plate grows and widens.
- The neural folds fuse along the midline forming the **neural tube**.
- The process by which the neural tube is formed from the neural plate is called primary **neurulation**



Fate of Neural Crest

- Differentiate into separate groups of neurons.

Group 1	Group 2	Group 3	Group 4
Sensory neurons of CN ganglia.	Autonomic ganglionic cells	Chromaffin cells of the adrenal medulla	Mesodermal cells alongside the neural tube - Somites
CN V, VII, IX and X of the head region.	Postganglionic neurons of the para and prevertebral ganglia of the SNS	Schwann cells- that forms myelin in PNS	Somites will develop into skeletal muscle, vertebrae and the dermal layer of the skin
Dorsal root ganglia, components of the body	Post ganglionic neurons of the PNS located in visceral organs	Melanocytes	

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Vesiculation/Segmentation

- The anterior aspect of the neural tube develops subdivisions which forms initially three and subsequently five brain vesicles.

Stage 3 Vesicle Primary	Stage 5 Vesicle Secondary
Prosencephalon - Forebrain	Prosencephalon- Telencephalon - Cerebrum - Diencephalon - Thalamus, Hypothalamus
Mesencephalon - Midbrain	Mesencephalon - Midbrain
Rhombencephalon - Hindbrain	Rhombencephalon- Metencephalon - Pons, Cerebellum Myelencephalon - Medulla
Caudally are cells from which the spinal cord develops.	The spinal cord develops from the posterior aspect.

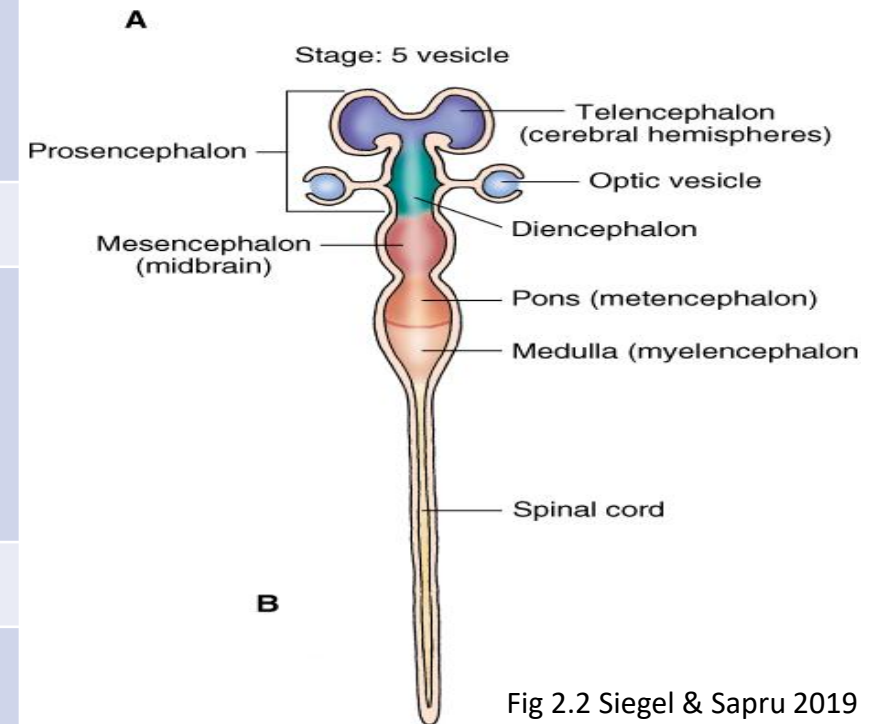
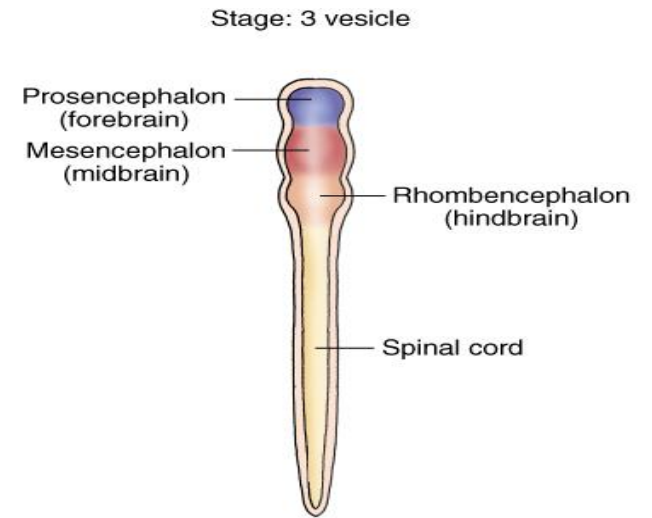


Fig 2.2 Siegel & Sapru 2019

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Forebrain Development/Diencephalon

Develops from swellings of the lateral aspects of the neural canal and mainly from the alar plates

Forms the **thalamus** dorsally and the **hypothalamus** ventrally

The thalamus displays the greatest amount of growth and is the largest component of the diencephalon. A small bridge is formed between its two sides the Massa Intermedia.

Other structures associated with the hypothalamus are the optic chiasm, the mamillary bodies, and the infundibulum.

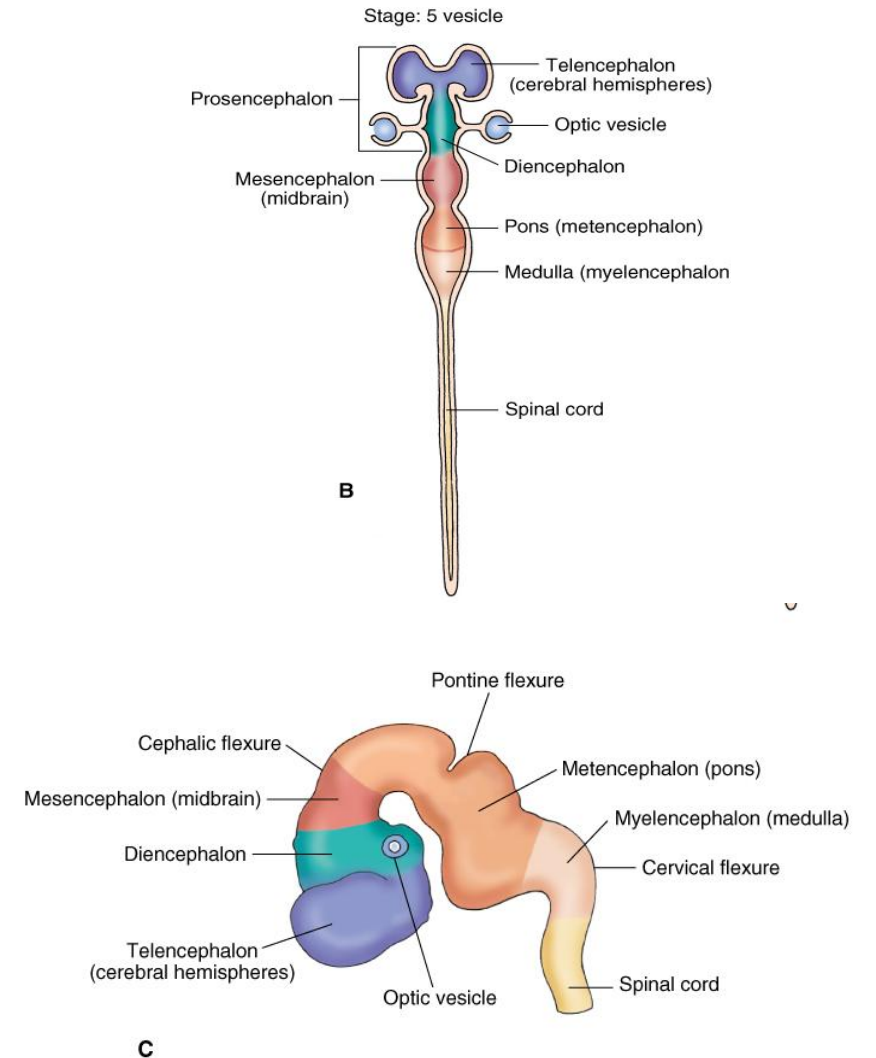


Fig 2.2 Siegel & Sapru 2019

Eye Development

- Week 4-5 the prosencephalon displays selective changes.
- Formation of **optic vesicle** at a ventral aspect of the anterior forebrain.
- Optic vesicle expands outwards while connected to the forebrain by the constricted **optic stalk**.
- The fiber bundles formed anterior to the optic chiasm are the **optic nerves**. Their continuation posterior to the chiasm is the **optic tract**.

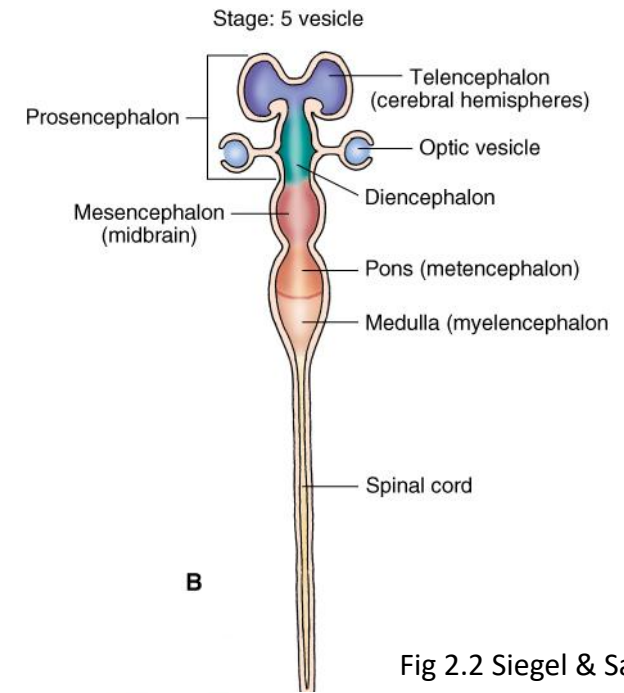
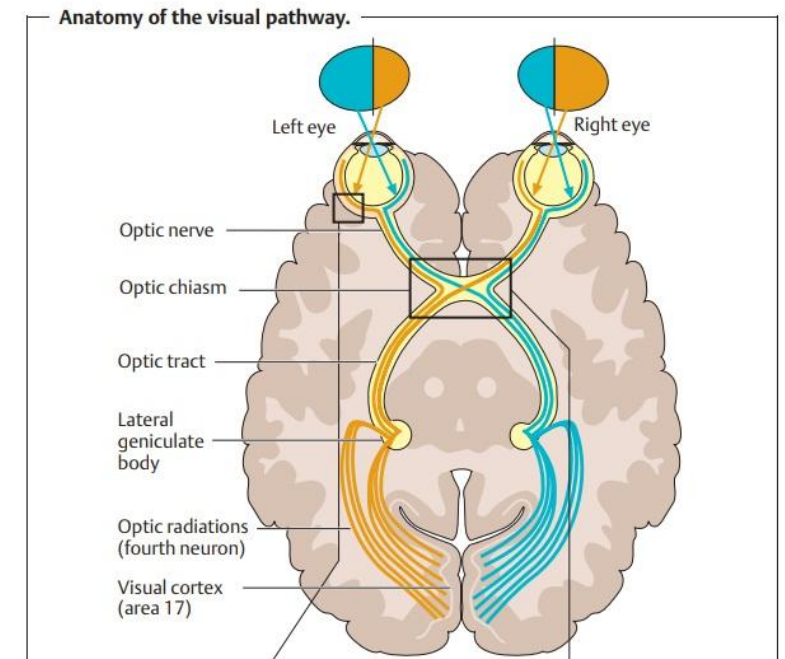
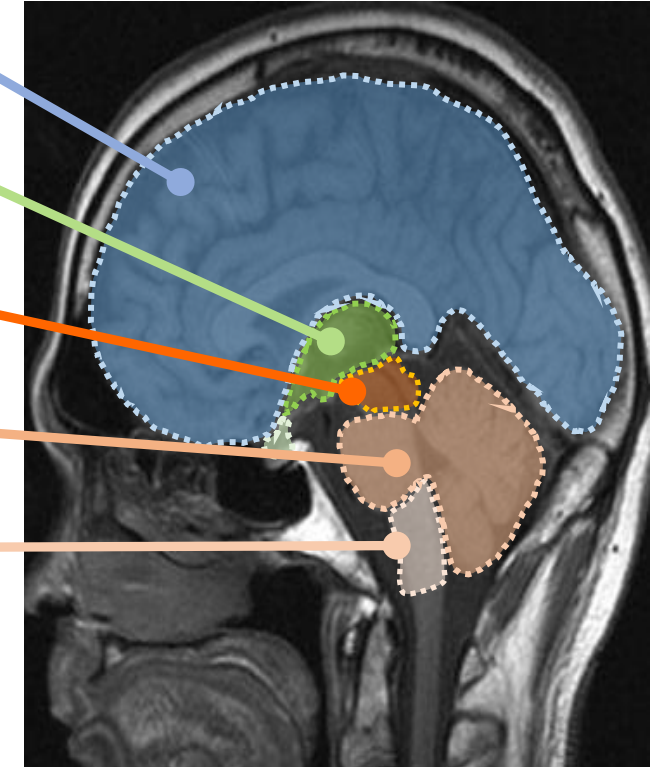
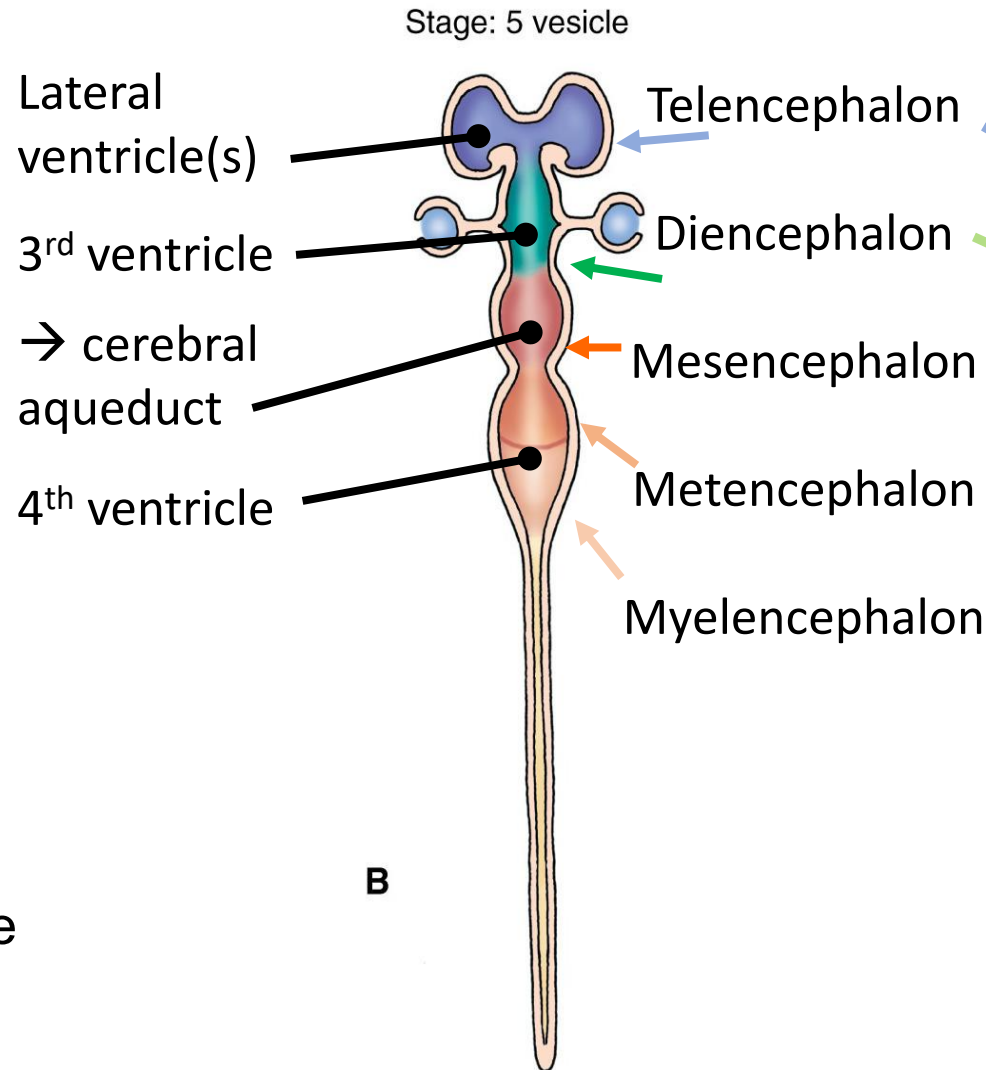


Fig 2.2 Siegel & Sapru 2019



Forebrain Development/Telencephalon

- Lies in front of the lamina terminalis
- Grows into the cerebral hemispheres from the dorsal aspect of the forebrain
- Sac-like structures within each hemisphere develop into the lateral ventricles.
- During month 3-4 the cerebral hemispheres continue to grow anteriorly and dorsally forming the frontal lobe
- It expands laterally and dorsally forming the parietal lobe.
- Posterior and ventral growth results in the development of the temporal and occipital lobes
- It contains the neocortex (six histologically distinct layers)



Left underlying image from: Siegel and Sapru, Essential Neuroscience © Wolters Kluwer **Figure 2.2B**

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Telencephalon Cont'd

- * The lateral ventricles are continuous with the third ventricle through the interventricular foramina (of Monroe).

- * Pyramidal cells are formed, and their axons form the internal capsule as well as the corpus callosum.

Granule cells are formed.

They receive input from regions of the thalamus.

Proliferation of immature neurons in the floor of the telencephalon forms the major components of the basal ganglia (caudate nucleus, putamen, globus pallidus)

Hippocampal formation from medial surface forms the fornix.

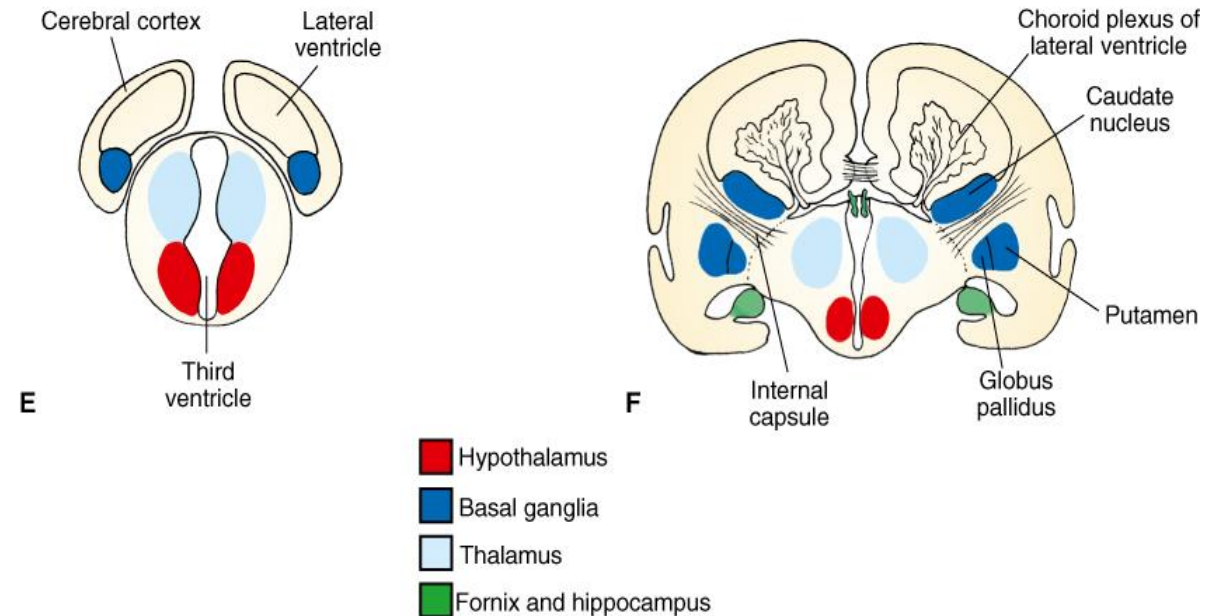


Fig 2.5 Siegel & Sapru 2019

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Spinal Cord

- The spinal cord consists of three layers:
- Ventricular layer- inner, most proliferative and the first to form.
- Mantle layer- intermediate and third layer to form. It contains the primary cell bodies of neurons and becomes the grey matter of the spinal cord.
- Marginal layer – second layer to form and the most outermost.
- The central canal is formed from the closure of the posterior neuropore.
- The sulcus limitans is formed along the lateral margin of the neural canal forming the walls. It is the shallow groove that separates the alar and basal plates of the mantle layer.

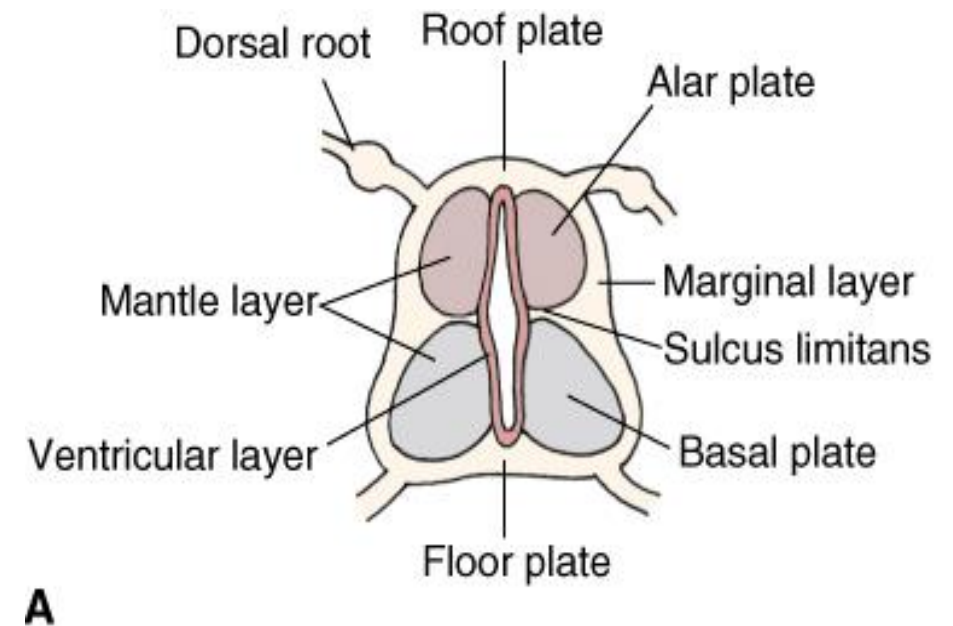


Fig 2.3 Siegel & Sapru 2019

Spinal cord : Alar & basal Plates

- Alar plates- formed from neurocytes that migrate dorsal to the sulcus limitans- contribute to sensory pathways.
- Basal plates- formed from neurocytes that migrate ventral to the sulcus limitans- contribute to motor pathways.
- Cells along the dorsal midline- roof plate
- Cells along the ventral midline- floor plate.
- Cells in an intermediate position adjacent to the sulcus limitans will become autonomic neurons.

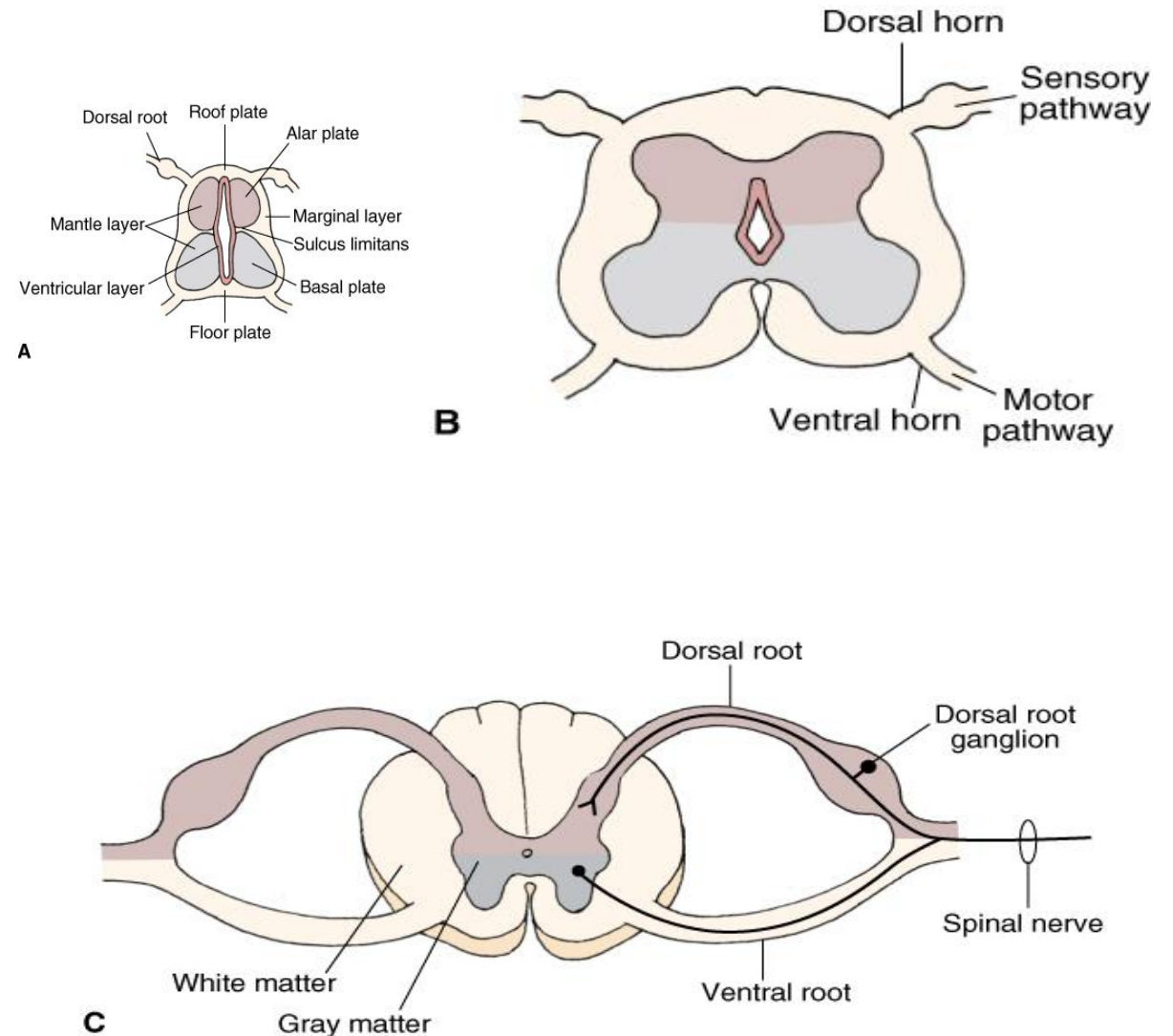


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Mesencephalon-Midbrain

- Three general regions:
- Tectal region - Dorsal
- Tegmental region - Intermediate
- Peduncular region - Ventral
- Neurocytes from the basal plate develop into motor neurons (CNIII, CNIV)

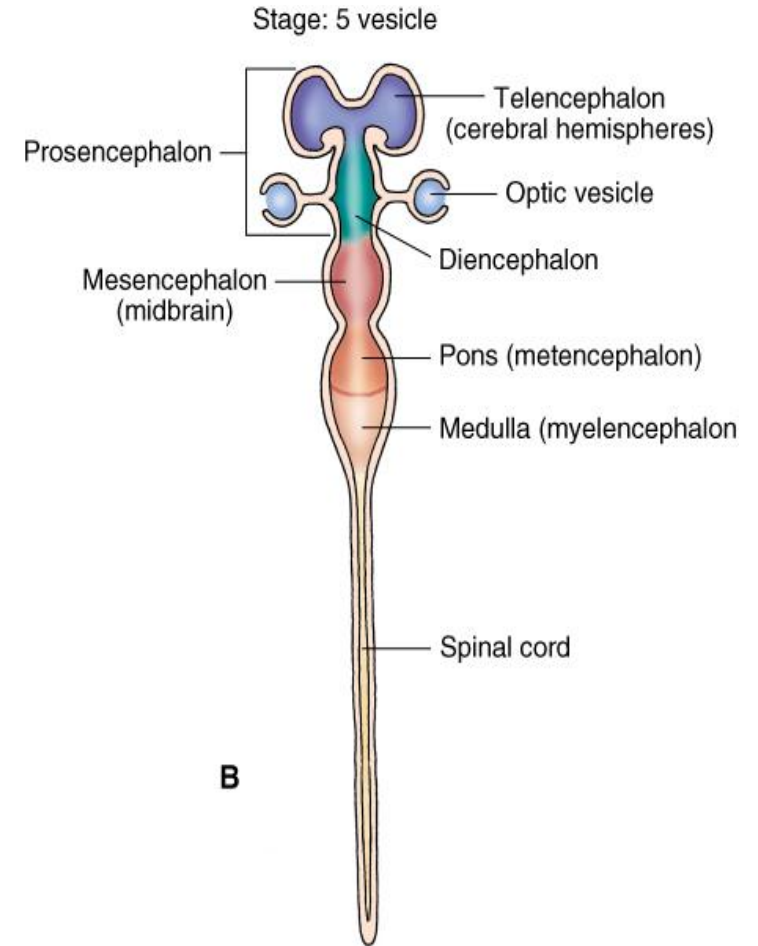


Fig 2.2 Siegel & Sapru
2019

Metencephalon-Pons & Cerebellum

- **Pons**

- Tegmentum – dorsally
- Basilar pons - ventrally

- **Cerebellum:** derived from the dorsal aspect of the alar plate.

- cells from the central region - vermis
- cells from the lateral region - cerebellar hemispheres
- Coordination of movement
- Spatial memory

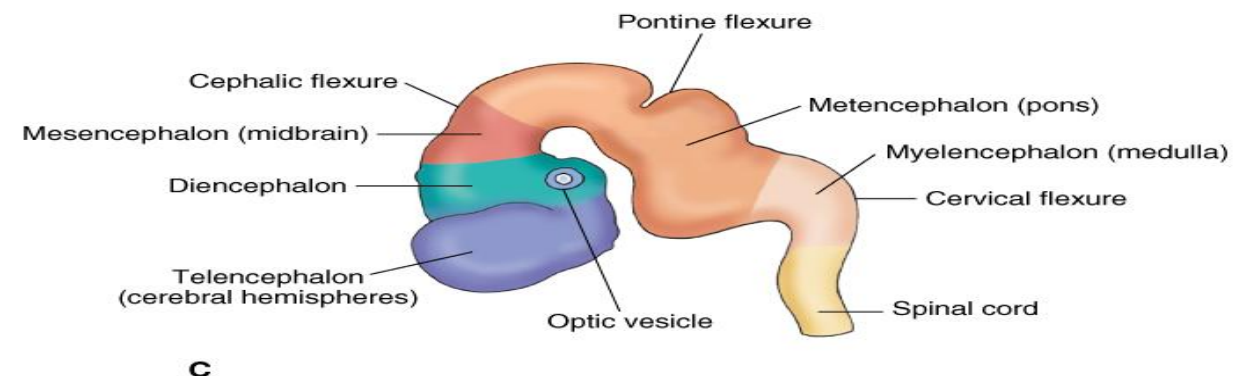


Fig 2.2 Siegel & Sapru 2019

Brainstem & Cerebellar Development

- Roof plate expands, alar plates located lateral to the basal plates
- Motor structures medial to sensory structures; GSE medial to GSA
- CNXII – Hypoglossal GSE – motor – innervates tongue
- CNVIII – vestibulocochlear special sensory afferent (SSA).
- Cerebellum formation (D,E): medial fusion of the rhombic lip, with vermis at midline

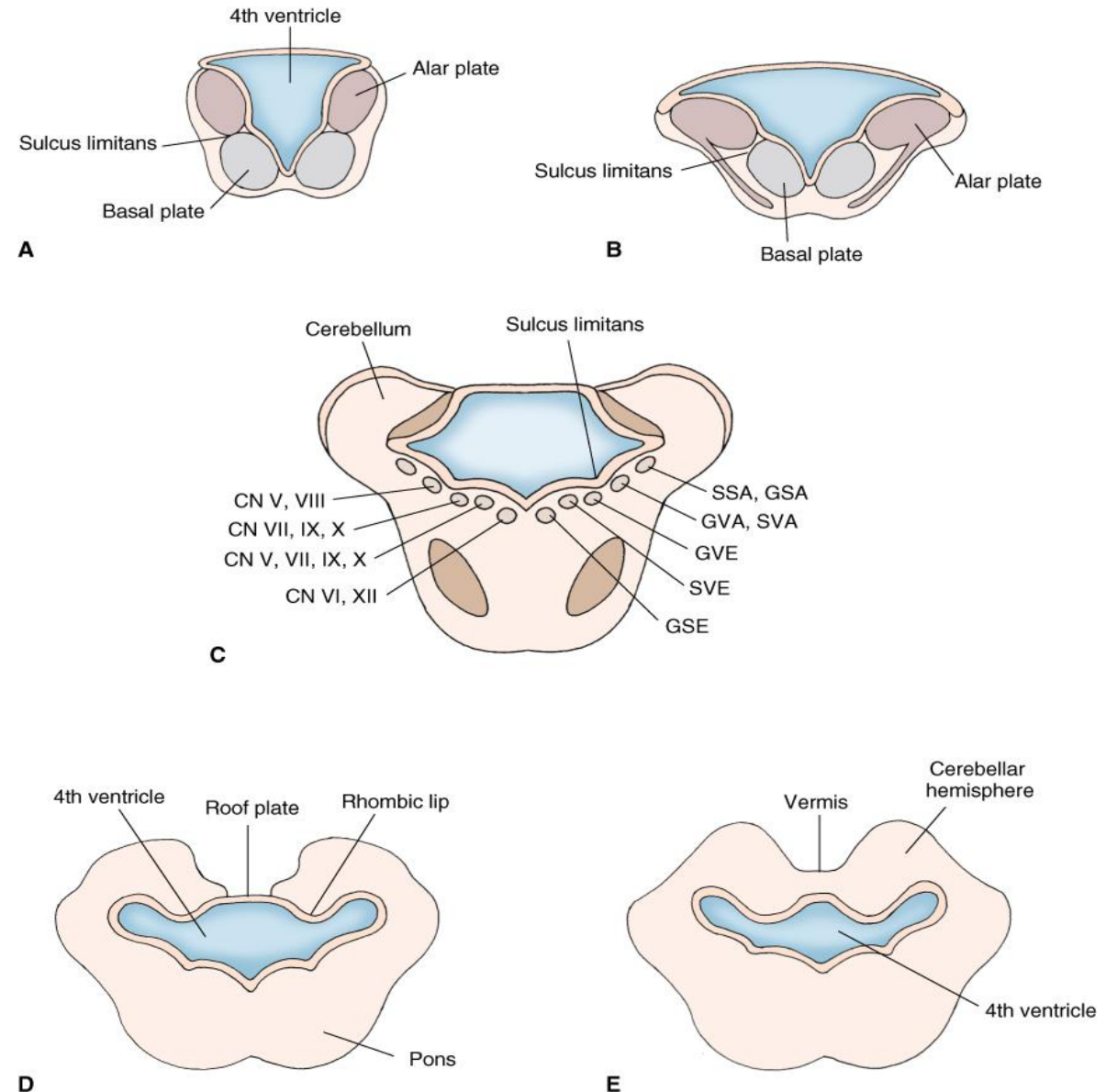
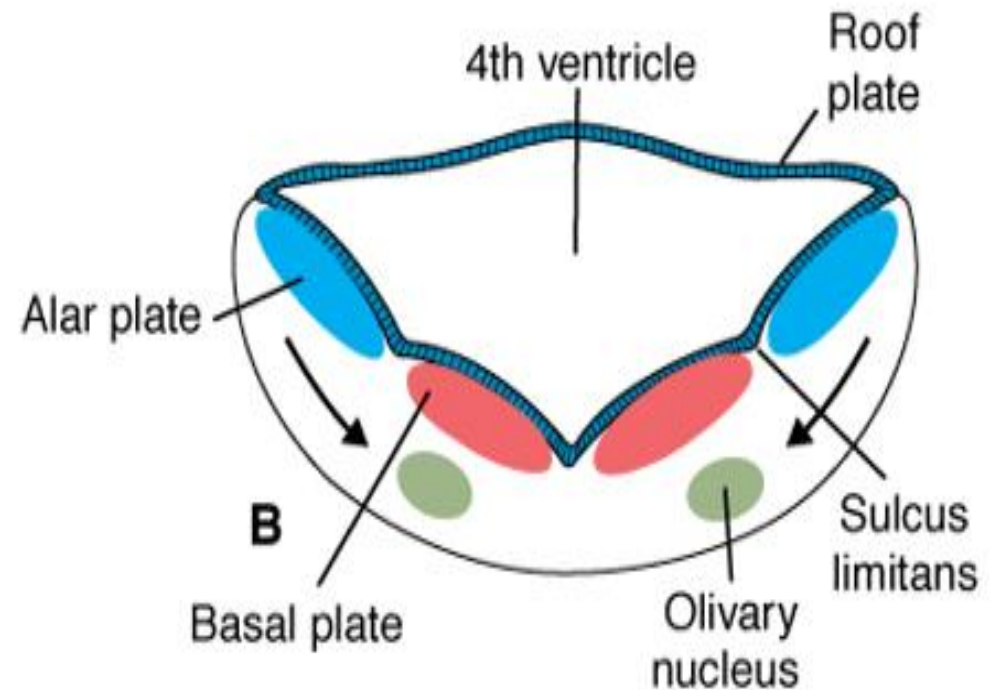


Fig 2.4 Siegel & Sapru 2019

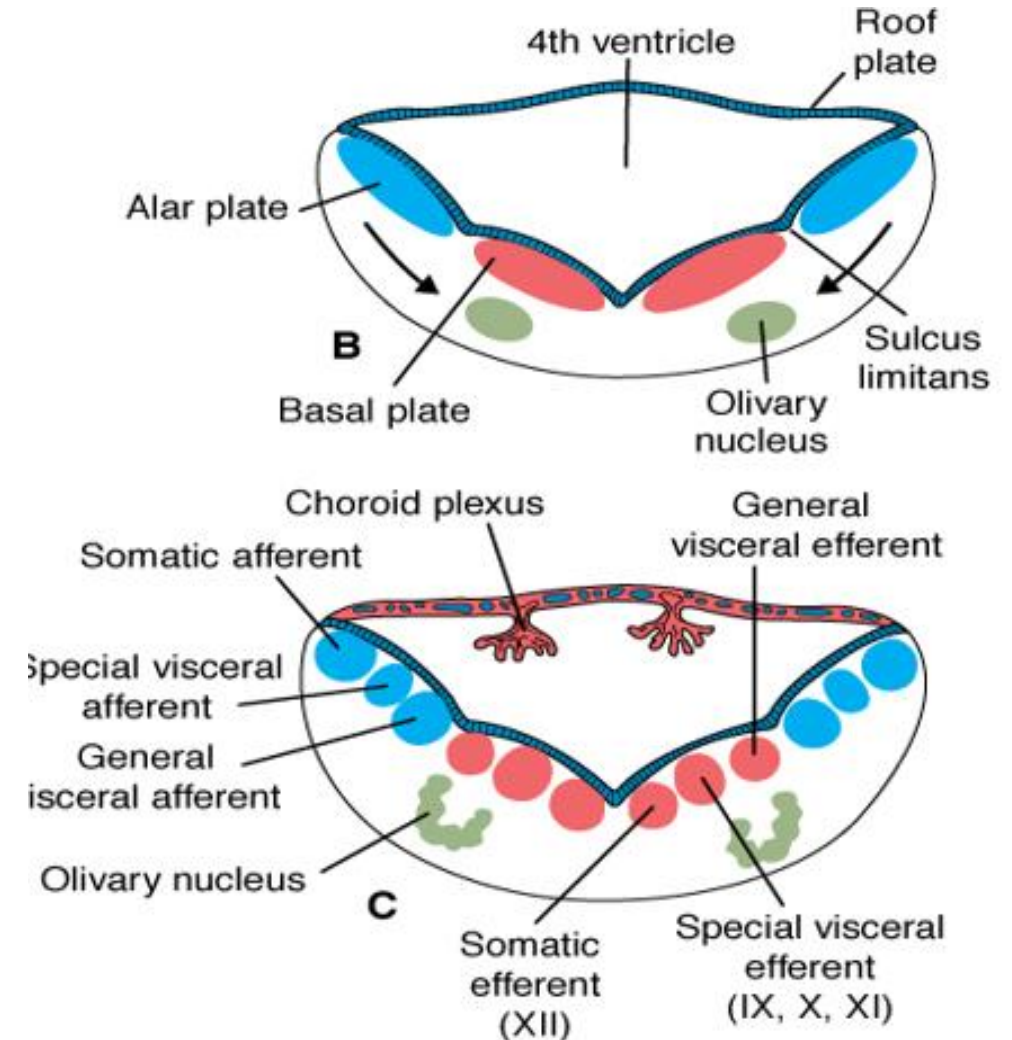
Myelencephalon-Medulla

- The neural canal expands to become the 4th ventricle, roof plate expands, and alar plate becomes lateral to basal plate.
- Motor and sensory neurons are arranged in columns in a medial to lateral fashion.
- Mesenchymal tissue in 4th ventricle becomes highly vascular, from the central portion of the roof it invaginates to become the choroid plexus.
- In the 4-6 month of development the foramen of Luschka (pair) and the foramen of Magendie (single) are formed.



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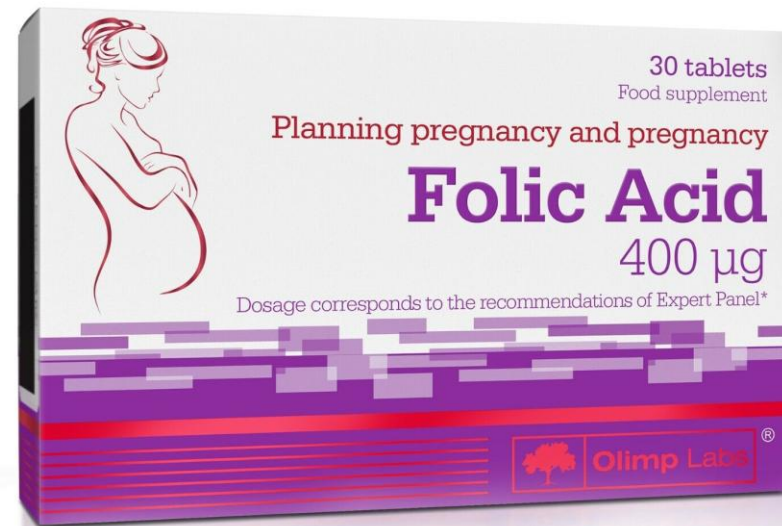
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Neural tube defects

Clinical

- Early development can be affected by a deficiency of folate which is critical for synthesis of various amino acids. A deficiency prevents appropriate cell turnover at the time of neural tube closure.
- To prevent neural tube defects folic acid or folic acid supplements are recommended.



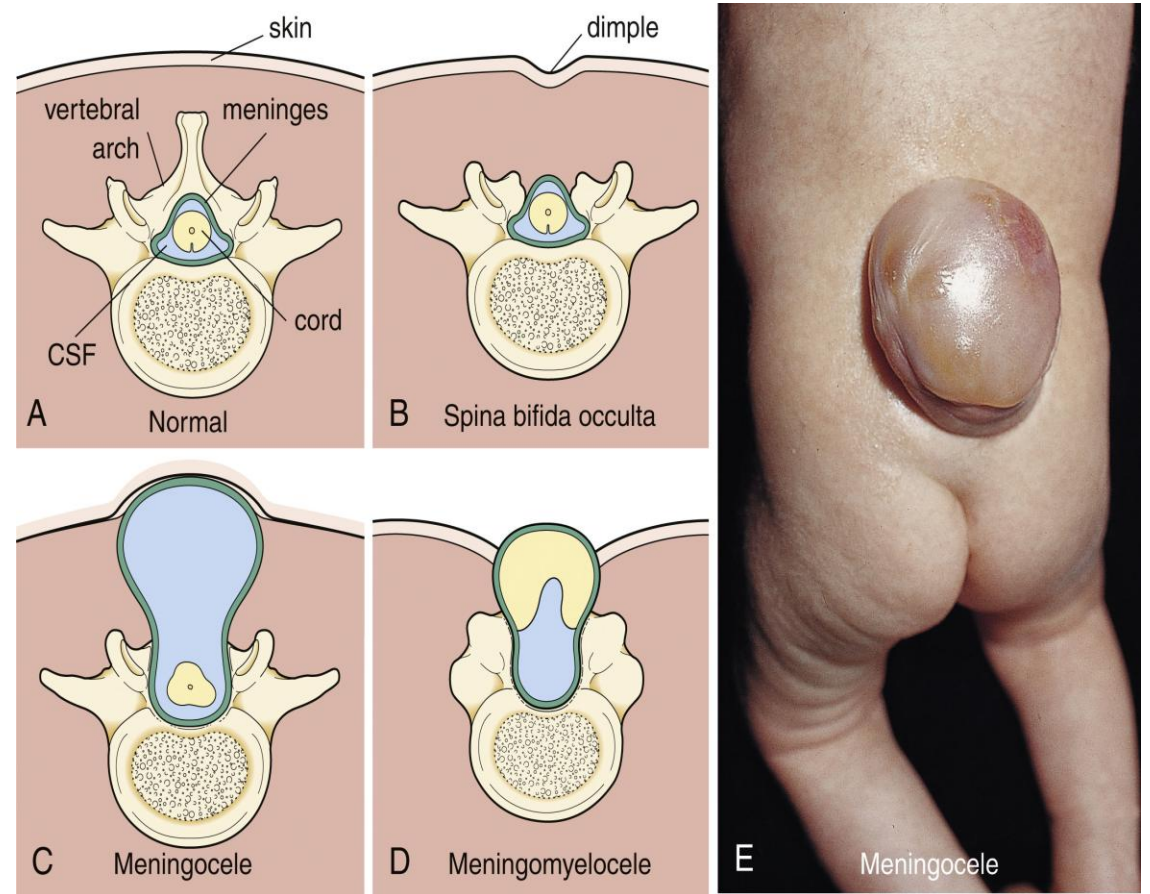
Spina Bifida-failure of vertebral canal closure

Spina Bifida Occulta

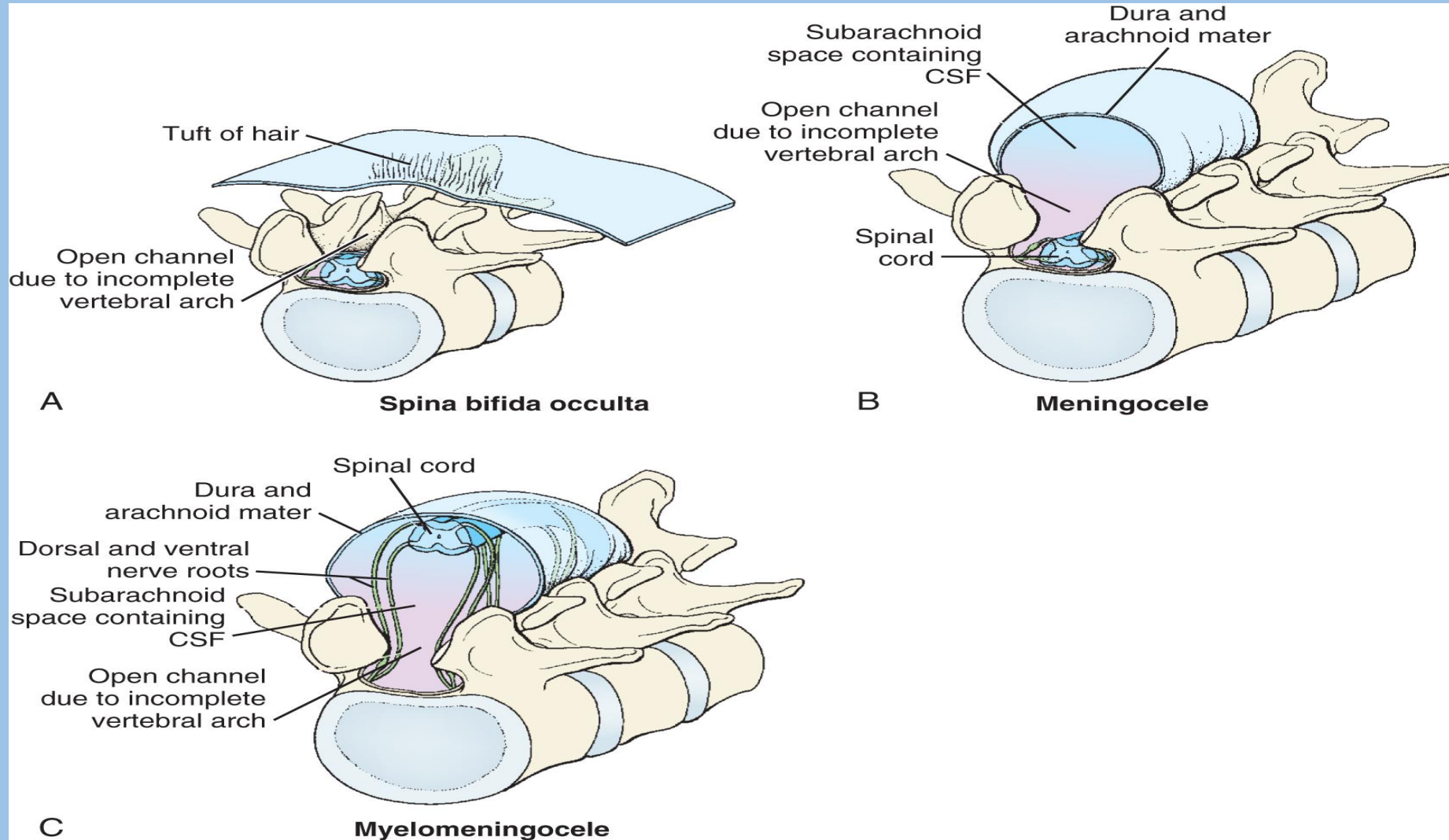
One or more vertebrae fail to close, no involvement of meninges or underlying spinal cord, overlying skin is closed.

Spina Bifida Aperta/Cystica

Protrusion of only meninges (meningocele) or spinal cord with meninges (meningo-myelocele)

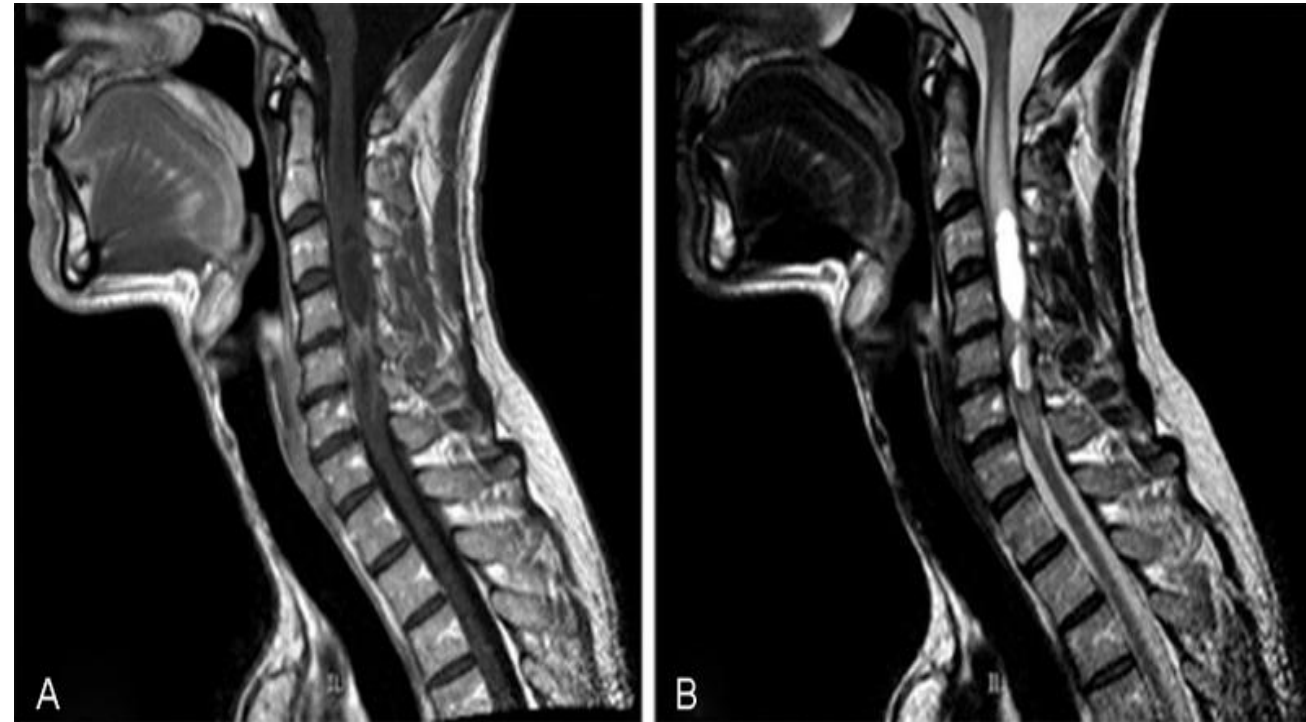


Spina Bifida



Syringomyelia

- A cavity formed in the region of the central canal. It is filled with CSF. It damages the crossing fibers of the spinothalamic tract resulting in segmental loss of pain and temperature.



Vandertop, W. P. (2014). Syringomyelia. *Neuropediatric*, 45(01), 003-009.

Tethered Cord Syndrome

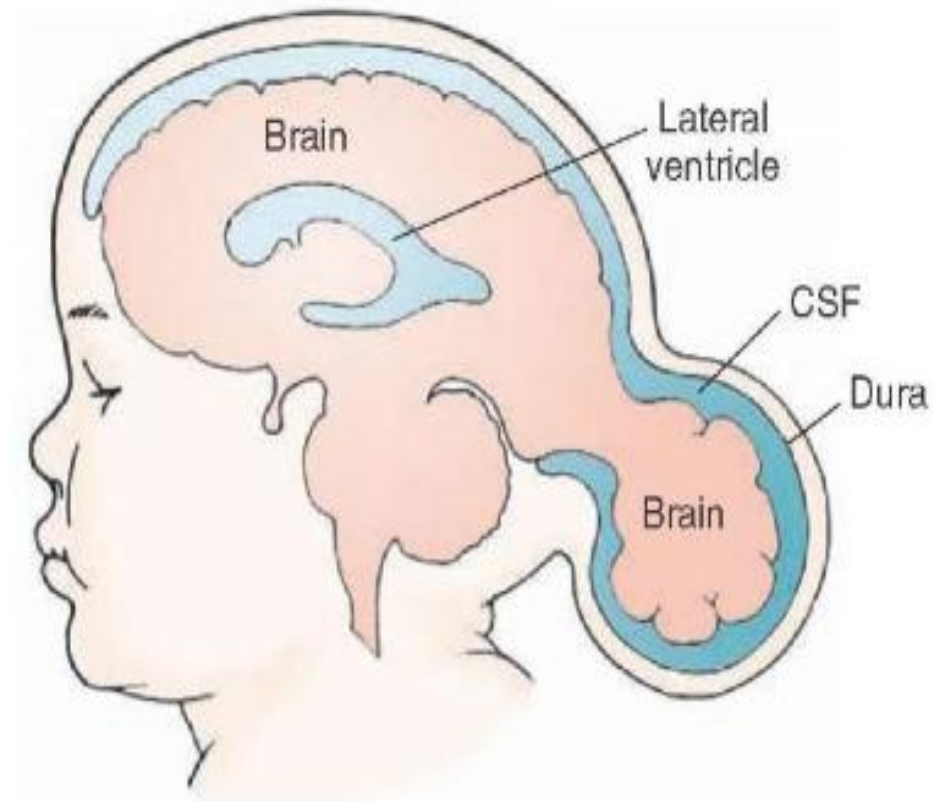
- Anchoring of lowest part of spinal cord to sacrum.



Walizai T, Rasuli B, Elfeky M, et al. Tethered cord syndrome. Reference article, Radiopaedia.org (Accessed on 24 Feb 2025) <https://doi.org/10.53347/rID-13494>

Encephalocele

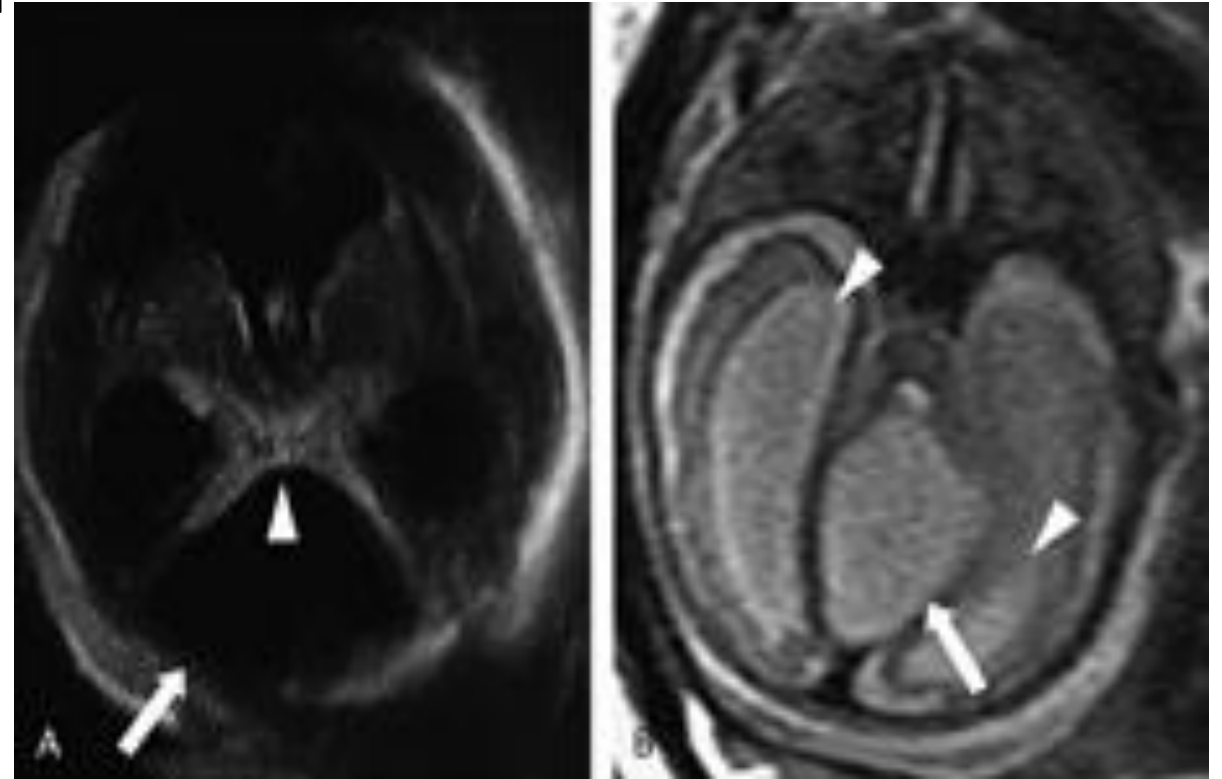
- Failure of portions of the anterior neural tube to close.



C Encephalocele

Dandy Walker Syndrome

- Congenital absence of the lateral apertures (foramen of Luschka) and the median aperture (foramen of Magendie).
- Results in complete/partial agenesis of the cerebellar vermis, cystic dilation of posterior fossa communicating with the fourth ventricle.



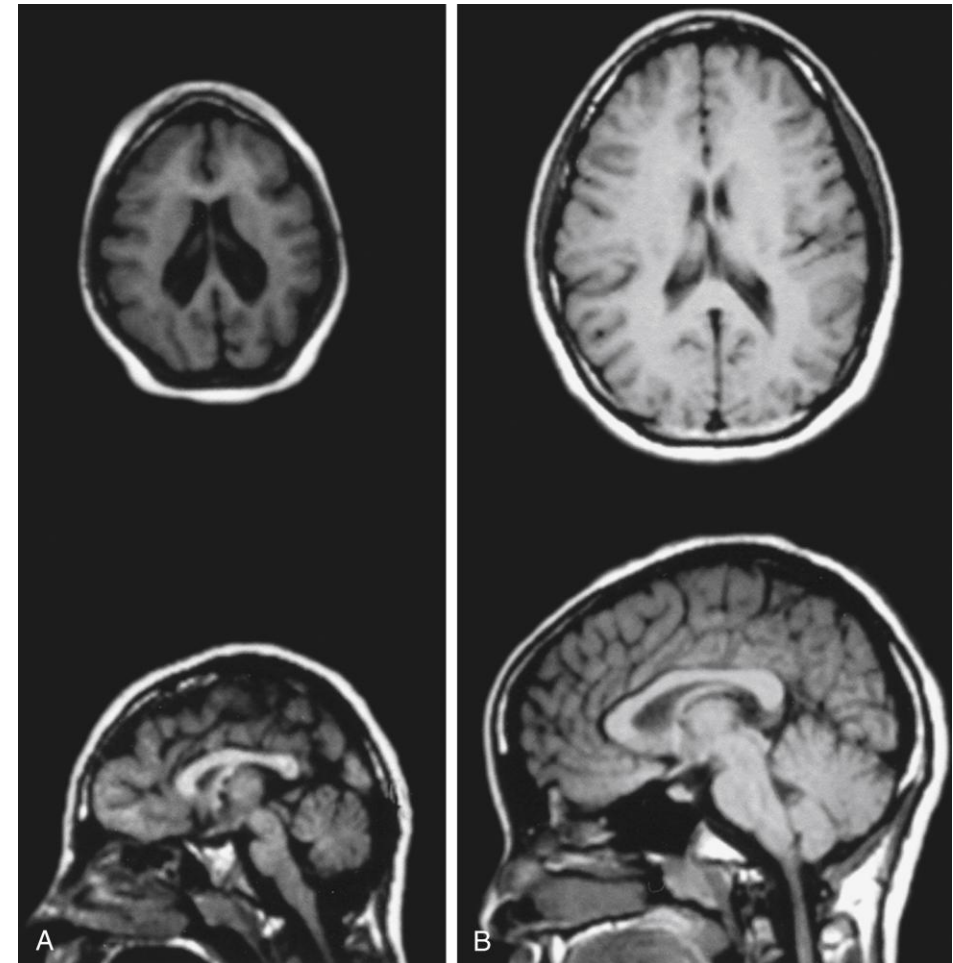
Correa, G. G., Amaral, L. F., & Vedolin, L. M. (2011). Neuroimaging of Dandy-Walker malformation: new concepts. *Topics in Magnetic Resonance Imaging*, 22(6), 303-312.

Anencephaly

- Partial or complete absence of the brain because of failure of the anterior neuropore to close.

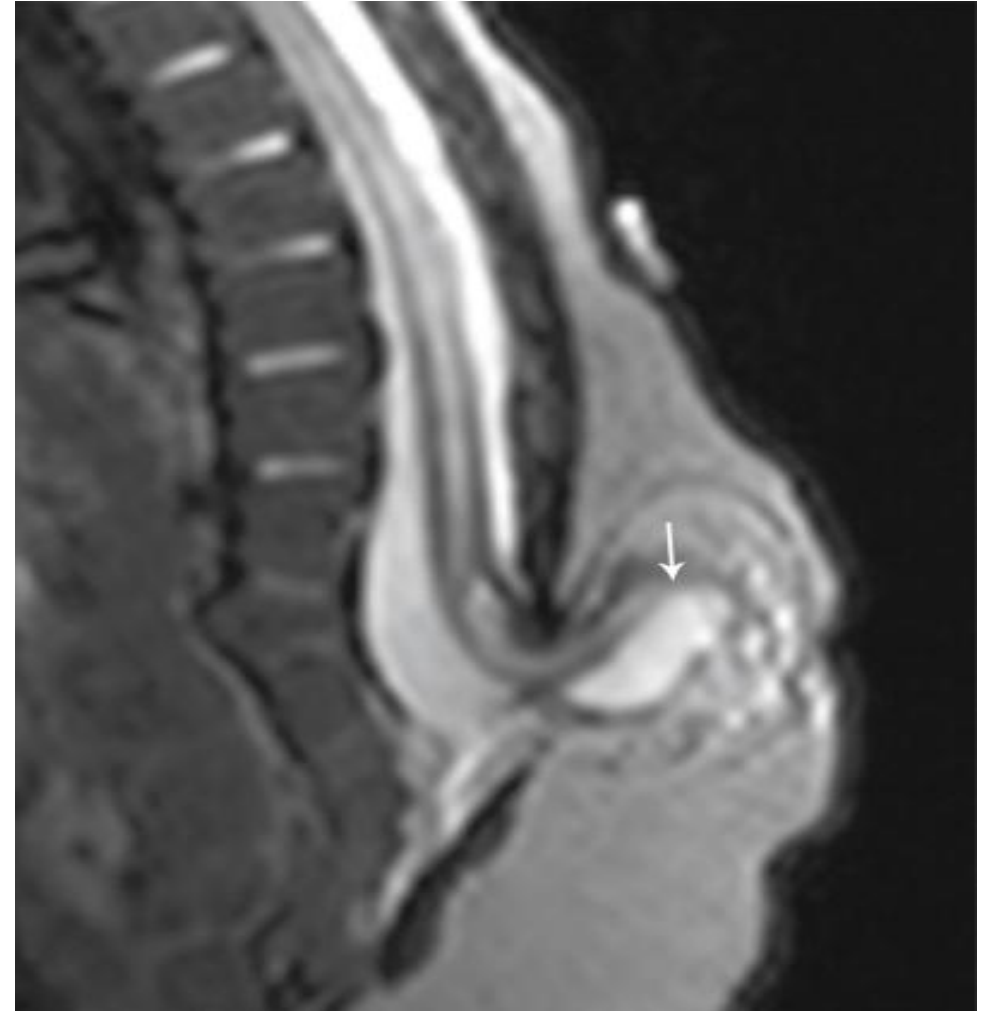
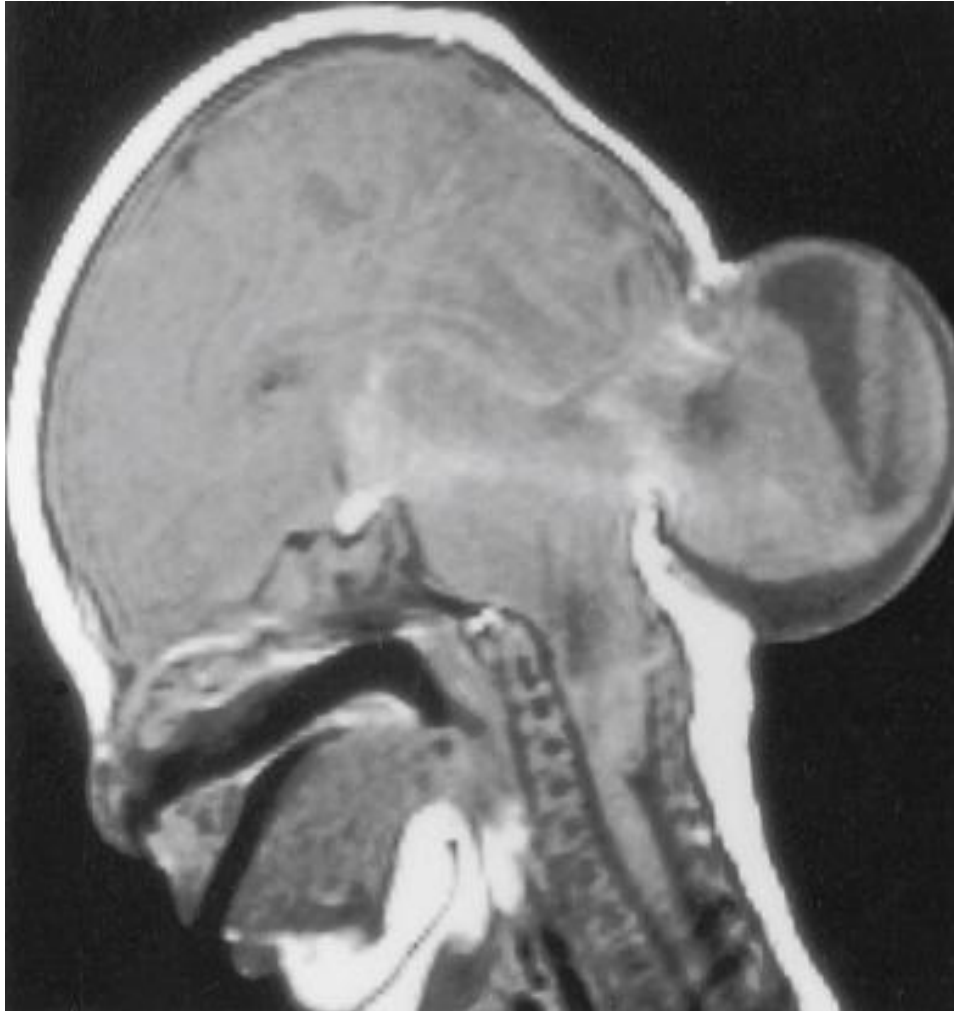


Courtesy of Dr. Roger Stevenson, Greenwood Genetic Center, Greenwood, SC. **B** Courtesy of Dr. David D. Weaver, Department of Medical and Molecular Genetics, Indiana University School of Medicine.

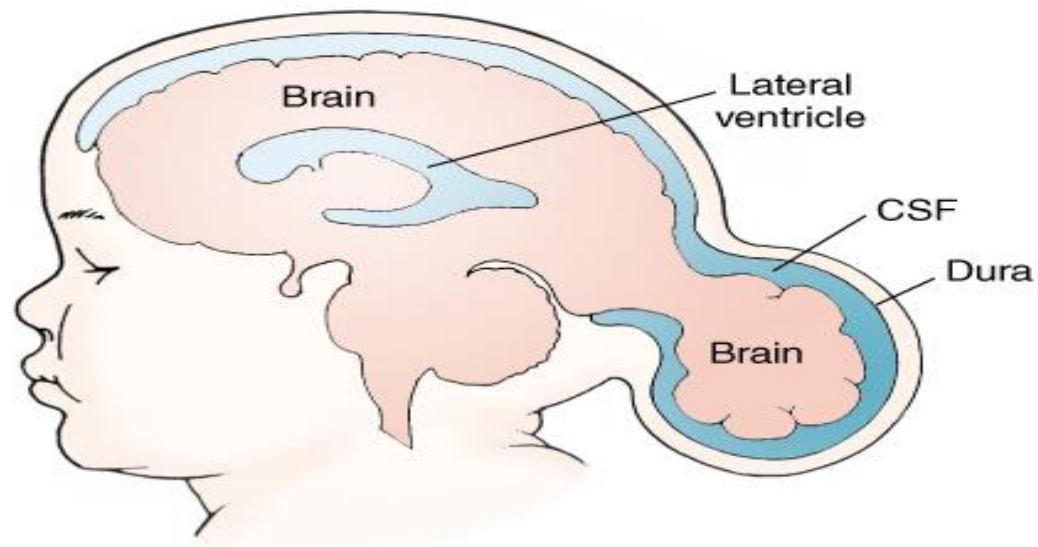
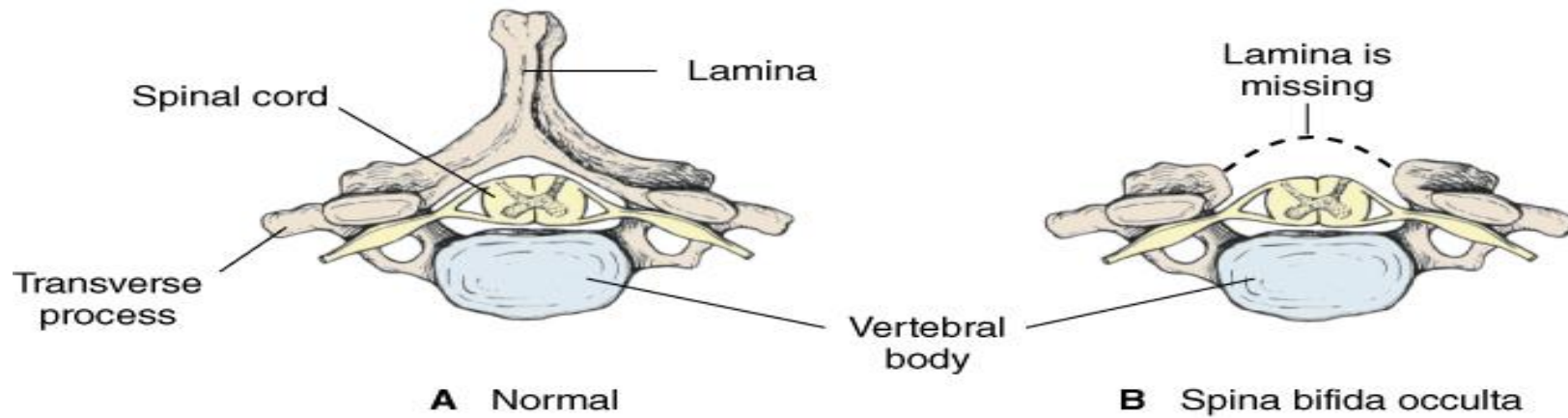


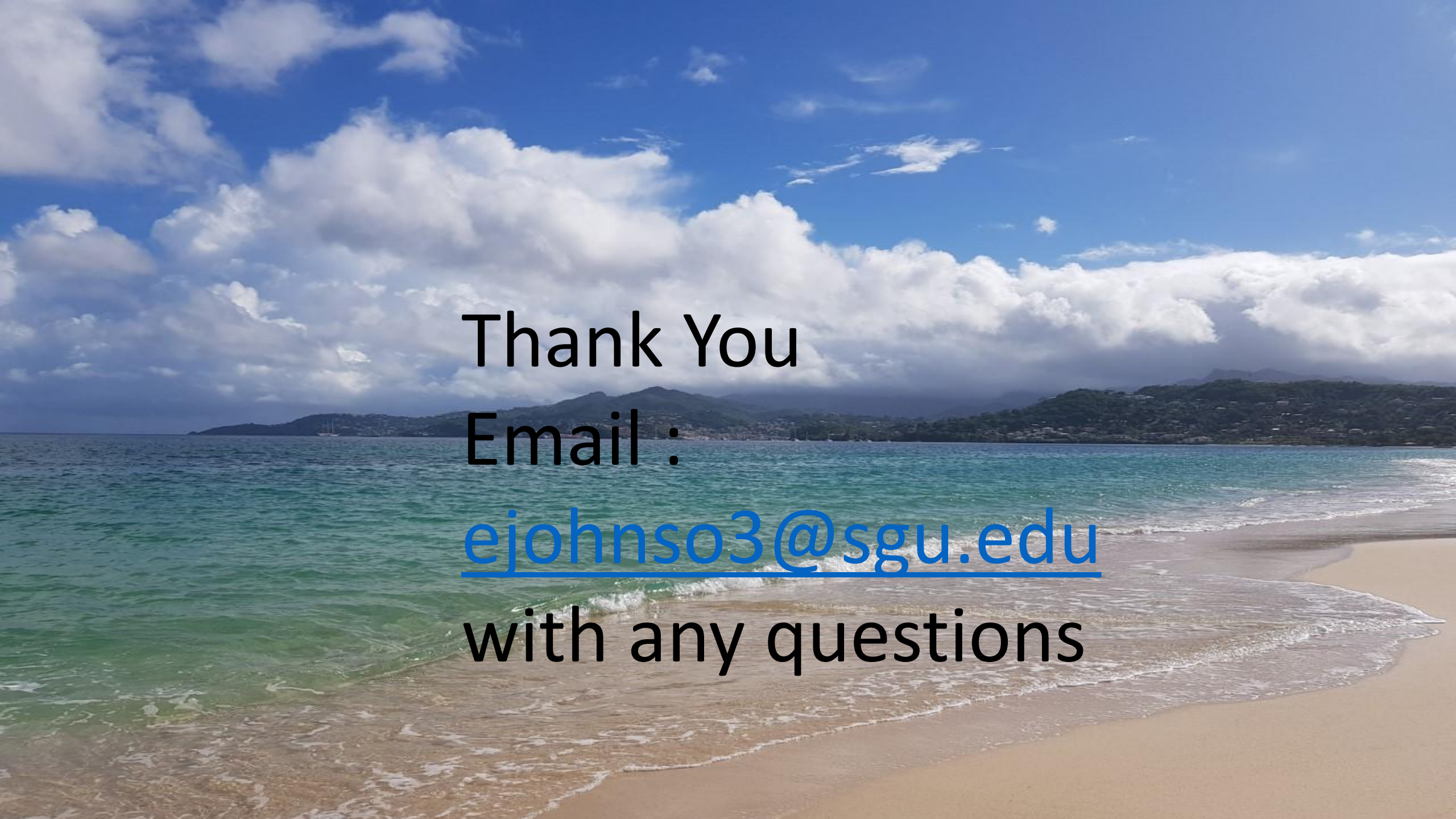
- *Larsen's Human Embryology*-Schoenwolf, Gary C., PhD; Bleyl, Steven B., MD, PhD; Brauer, Philip R., PhD; Francis-West, Philippa H., PhD. © 2021.

MRI depicting a cephalocele & a meningocele



Summary-Brain and Spinal Cord Abnormalities





Thank You

Email :

ejohnso3@sgu.edu

with any questions